

Security and Safety Engineering Qualitative Assessment Report

ANONYMOUS ID - 27

EXECUTIVE SUMMARY (max 150 words)

This report assesses cybersecurity risks associated with the proposed Smart Court remote-working solution for Civil and Small Claims Court clerks at the Tribunal of Zaltbommel. The assessment identified two critical primary assets: Court Case Information and Court Proceedings and Deadlines, where compromise could result in information disclosure, unauthorised record changes, operational disruption, and loss of public trust.

The key supporting assets identified were court-managed laptops, specialised judicial software, VPN infrastructure, user credentials, and network infrastructure. The highest-priority threats include ransomware, malware infection, phishing, credential theft, laptop loss, and network lateral movement caused by insufficient internal segmentation. These risks are increased by remote working, password-sharing practices, and weaknesses in internal network separation.

Recommended controls include MFA, EDR, patch management, offline backups, security awareness training, least-privilege access, and internal firewall segmentation to reduce cybersecurity risks and support secure court operations.

Work submitted in partial fulfillment for the course of Security and Safety Engineering – Vrije Universiteit Amsterdam - a.a. 2024/2025

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Threats Assessment Report

Rebuttal To Reviewers

1. Assets and Threats Previous Reports

The following comments on the sections have been received for the threat assessment individual reports:

1. On the abstract and the summary of the findings:

- * The executive summary and findings were considered too generic and did not sufficiently reference the specific Smart Court case study.
- * The summaries did not include key threat assessment results, impact scores, or the most significant threats.
- * Asset naming was inconsistent, as Court Case Records, Legal Documents, and Judicial Decisions were still listed separately despite being consolidated.

2. On the target of evaluation and assets:

- * The scope and boundaries of the assessment were not clearly defined.
- * The distinction between Civil and Small Claims Courts (in scope) and Penal/Prosecution Courts (out of scope) was not explicitly stated.
- * Several primary assets overlapped and should be consolidated to avoid redundancy.
- * More case-study-specific supporting assets and technical details were requested.

3. On threat evaluation:

- * Threats were considered generic and not sufficiently tailored to the Smart Court architecture.
- * Important threats such as password sharing, remote access software misuse, household access to devices, and cross-court network compromise due to the lack of internal firewalls were omitted.
- * Threat descriptions required more detailed attack scenarios and stronger links to the case study.

4. On vulnerabilities:

- * Vulnerabilities were described using generic terms rather than specific weaknesses identified in the case study.
- * The absence of internal network segmentation, remote management software exposure, and clerk behaviour assumptions should be reflected more explicitly.

5. On threat impact:

- * Impact justifications required stronger links to the case study.
- * Confidentiality impacts should consider the absence of internal firewalls and possible exposure of sensitive court information.
- * Availability impacts should explicitly discuss missed statutory deadlines and invalidated proceedings.

6. On likelihood:

- * Likelihood scores required more detailed justification.
- * Several likelihood factors were not clearly connected to the SecRAM methodology or the Smart Court environment.
- * Some scoring interpretations required correction.

7. On links between threats, vulnerabilities, and assets:

- * The relationships were generally clear but required more detailed explanations showing how supporting assets affect primary assets.
- * Generic mitigating controls should not be assumed unless explicitly stated in the case study.

8. On supporting assets:

- * Some direct/indirect asset relationships required clarification.
- * Recommended adding Network Infrastructure.

9. On other sections of the report:

- * Remaining template instructions should be removed.
- * Formatting improvements were suggested, including colour consistency in tables and clearer presentation of threat impacts.

10. General consistency:

- * All references to consolidated assets should remain consistent throughout the report.
- * Rebuttal logs should be fully completed and clearly explain all implemented changes.

The remaining comments have been addressed as follows:

- a. Court Case Records, Legal Documents, and Judicial Decisions have been consolidated into a single primary asset called "Court Case Information" to eliminate redundancy and improve consistency.
- b. The Target of Evaluation now explicitly defines the scope as the remote-working solution for Civil and Small Claims Court clerks, while excluding Penal and Prosecution Court operations.
- c. Executive Summary and Summary of Findings sections have been revised to include key impact results, major threats, and the most critical assets identified during the assessment.
- d. Threat scenarios and vulnerabilities have been rewritten to reference Smart Court-specific weaknesses, including password sharing, lack of internal firewalls, remote access software exposure, and home-working risks.
- e. Likelihood and impact justifications have been expanded with case-study-specific explanations and clearer links between supporting assets, vulnerabilities, threats, and primary assets.
- f. Added Network Infrastructure as a supporting asset and adjusted report as such.

The original reports have been merged as follows:

- g. Asset assessment findings were used as the foundation for the threat assessment by carrying forward the validated primary and supporting asset structure.
- h. Consolidated asset definitions and updated impact assessments were applied consistently throughout both reports.
- i. Threat scenarios were mapped directly to the revised asset model to ensure consistency between the asset and threat assessments.
- j. Supporting asset relationships and scope assumptions from the asset report were integrated into the threat assessment to improve traceability and reproducibility.
- k. Executive summaries, findings, terminology, and asset naming conventions were aligned across both reports to produce a single coherent and consistent risk assessment.

Risk Assessment Report - Complete

ANONYMOUS ID - 27

2. TARGET OF EVALUATION (max ½ page)

The Target of Evaluation (ToE) is the proposed Smart Court remote-working solution used by Court Clerks within the Civil Court and Small Claims Court divisions of the Tribunal of Zaltbommel. The assessment focuses on the systems, technologies, processes, and information assets required for clerks to remotely access, process, manage, and protect court information while supporting judicial activities.

The primary assets within scope are:

1. **Court Case Information** – Includes electronic case files, case records, evidence, litigant information, pleadings, submissions, correspondence, procedural documents, and other case-related data managed throughout the lifecycle of a court case.
2. **Court Proceedings and Deadlines** – Includes hearing schedules, filing deadlines, draft judgments, final rulings, court orders, procedural timelines, judicial decisions, and documentation required to conduct court proceedings.

The supporting assets evaluated within the ToE include court-managed laptops, specialised judicial case management and document management software, VPN infrastructure, user authentication credentials, and network infrastructure consisting of firewalls, VLAN segmentation, Layer 2 switching, and internal connectivity between tribunal systems. These assets are included because their compromise could affect the confidentiality, integrity, or availability (CIA) of the primary assets.

The assessment follows a risk-based approach where assets that store, process, transmit, or provide access to critical information are analysed to identify potential cybersecurity risks [1]. The objective is to evaluate threats, vulnerabilities, and impacts associated with the Smart Court remote-working environment, particularly risks related to endpoint security, authentication, remote access, specialised applications, and network architecture.

The assessment considers the increased attack surface introduced by remote working outside the tribunal's physical security boundaries. It assumes that some clerks may share passwords through external messaging applications such as WhatsApp, contrary to security policies, and that the President of the Court reviews only high-level judicial software logs without routinely analysing VPN logs, IP logs, endpoint monitoring data, or network-level audit records. These assumptions are included to assess weaknesses in identity management, monitoring capabilities, and internal network segmentation, which may enable unauthorised access, delay incident detection, or allow attackers to move between connected court environments [2].

3. SUMMARY OF FINDINGS (max ½ page)

The cybersecurity assessment of the proposed Smart Court remote-working solution for Civil and Small Claims Court clerks at the Tribunal of Zaltbommel identified two critical primary assets: Court Case Information and Court Proceedings and Deadlines. These assets were assessed against the confidentiality, integrity, and availability (CIA) security objectives. A compromise of confidentiality could expose sensitive litigant information and judicial documents, while loss of integrity or availability could result in unauthorised changes to case records, disrupted court operations, delayed hearings, and missed statutory deadlines.

The supporting assets identified were court-managed laptops, specialised judicial software, VPN infrastructure, user credentials, and network infrastructure. The highest-priority risks included ransomware attacks, malware infections, phishing, credential theft, and network lateral movement. These risks were increased by remote working practices, password-sharing assumptions, and the lack of internal firewall separation between Civil, Small Claims, Penal, and Prosecution networks.

To mitigate these risks, recommended controls include Multi-Factor Authentication (MFA), Endpoint Detection and Response (EDR), application whitelisting, patch management, offline backups, security awareness training, least-privilege access controls, internal firewall segmentation, network access controls, and enhanced monitoring. These measures significantly reduce the likelihood of successful exploitation, although network segmentation requires continued improvement. Overall, protecting the confidentiality and integrity of Court Case Information remains the highest priority due to its impact on judicial fairness, legal compliance, and public trust.

4. RISK ANALYSIS (max 7 pages)

3.1. Impact Assessment (Step 1)

At Step 1, we identified the primary assets and assessed their impact (see Tables 1.1 and 1.2). The main primary assets identified in Table 1.1 are **Court Case Information** and **Court Proceedings and Deadlines**. These assets are critical to the operation of the courts because they directly support case management, judicial decision-making, and the timely delivery of justice. The effective protection and operation of these assets are essential to ensuring that legal proceedings are conducted fairly, accurately, and within statutory timeframes.

The confidentiality, integrity, and availability (CIA) of these assets are important because they represent the fundamental security objectives used to evaluate information security risks. Protection of these properties ensures that sensitive information remains accessible only to authorised users, remains accurate and trustworthy, and is available when required for operational activities [2].

For **Court Case Information**, confidentiality is essential because court records, legal documents, evidence, contracts, complaints, and judicial decisions often contain highly sensitive personal, legal, and procedural information. Unauthorised disclosure could violate privacy obligations, expose confidential information relating to litigants, witnesses, judges, and court staff, and undermine confidence in the judicial system. Protection of such information is a key requirement of information security management frameworks such as ISO/IEC 27001, which emphasises the safeguarding of information assets through appropriate confidentiality controls [5]. Integrity is equally important because any unauthorised modification of case records, evidence, legal documents, or judicial decisions could compromise the accuracy of proceedings, affect judicial outcomes, and potentially lead to wrongful decisions, appeals, retrials, or legal disputes. Availability is also critical because authorised personnel must be able to access case information when required throughout the judicial process. Prolonged unavailability could delay hearings, reduce operational efficiency, and affect compliance with legal deadlines.

For **Court Proceedings and Deadlines**, availability is particularly critical because courts must process cases, conduct hearings, and issue judgments within legally mandated timeframes. Disruption of these services could delay hearings, prevent courts from meeting statutory obligations, obstruct the administration of justice, create case backlogs, and negatively affect individuals and organisations awaiting legal outcomes. Integrity is also highly important because court processes, schedules, and deadlines must remain accurate and reliable to ensure proceedings are conducted correctly. Any unauthorised alteration of procedural information could result in missed deadlines, scheduling conflicts, procedural errors, and unfair outcomes. Confidentiality remains relevant but has a comparatively lower impact because many court proceedings are ultimately intended to become public. Nevertheless, premature disclosure of sensitive case details or restricted procedural information could still negatively affect ongoing proceedings and certain participants.

This assessment focuses primarily on the identified information and service assets because they are directly involved in judicial proceedings and the administration of justice. Assets such as office equipment, facilities, and general administrative resources were not considered primary assets because they provide indirect support to court operations rather than directly contributing to judicial decision-making and case resolution.

As shown in Table 1.2, confidentiality, integrity, and availability are all relevant security considerations for the identified primary assets. **Court Case Information** exhibits high impact across all three security properties, reflecting the sensitivity and operational importance of judicial records and legal documentation. For **Court Proceedings and Deadlines**, integrity and availability have the highest impact because accurate and timely processing of cases is essential to the administration of justice, while confidentiality has a comparatively lower impact as many court proceedings eventually become public. This assessment follows the SecRAM approach, where asset impact is evaluated according to the consequences of compromising security properties and their effect on organisational objectives [1]. The highest impacts were associated with compromises to the integrity and availability of both primary assets, and to the confidentiality of Court Case Information. Such compromises could invalidate court proceedings, delay or obstruct justice, breach legal and regulatory obligations, damage public confidence in the judicial system, and negatively affect the rights and interests of individuals and organisations relying on the courts [4].

Table 1.1: Primary Asset (PA) Identification

1.1 Primary Asset (PA) Identification			
Primary Asset ID	Primary Asset Description	Type (information or service)	Justification
Court Case Information	Court case records, legal documents (contracts, leases, complaints), and	Information	This is the core information asset that supports all court operations. It contains evidence, legal documentation, case details, and final judgments. Unauthorised disclosure, modification, or loss of this

	judicial decisions used by clerks and courts throughout legal proceedings.		information could compromise court decisions, violate confidentiality, and undermine public trust in the justice system
Court Proceedings and Deadlines	The processes and services involved in hearing, managing, and resolving court cases within legally mandated timeframes.	Service	The court must process cases efficiently and meet statutory deadlines. Disruption of this service could delay justice, obstruct legal proceedings, and negatively impact individuals and organisations relying on timely court decisions.

3.2. Supporting Asset Identification & Valuation (Step 2)

At Step 2, supporting assets related to the primary assets identified in Step 1 were identified and analysed (see Table 2.1). The primary assets, Court Case Information and Court Proceedings and Deadlines, depend on several supporting assets to ensure secure access, processing, storage, and timely management of judicial information. The supporting assets identified include Laptops, Specialised Software, VPN, User Credentials, and Network Infrastructure. These assets provide the technological capabilities required for court personnel to access judicial systems remotely, manage case-related information, and support court operations within legally mandated deadlines.

These supporting assets enable authorised court personnel to access court systems remotely and perform case-management activities while maintaining connectivity with internal tribunal resources. The proposed Smart Court solution increases the attack surface by extending access beyond the controlled physical environment of the tribunal building. Court-managed laptops, remote access technologies, authentication mechanisms, and the underlying network architecture introduce additional pathways that could be exploited by attackers. Cybersecurity frameworks emphasise the importance of identifying and protecting supporting assets because their compromise may provide attackers with access pathways to critical information systems and essential services [2][5].

Laptops provide the endpoint devices through which court clerks access, view, update, and process court information. Since these devices are used outside the secure court environment, they introduce risks related to theft, malware infection, and unauthorised physical access. Specialised software supports the management of court case records, legal documents, workflow states, judicial decisions, and scheduling information required for judicial operations. The VPN (Juniper Junos Pulse) provides remote connectivity between court-managed laptops and internal court systems, enabling authorised remote access while protecting communications. User credentials provide authentication and authorisation mechanisms that control access to court systems and ensure that only authorised personnel can access protected resources [3].

Network Infrastructure was also identified as a critical supporting asset because it provides the communication foundation connecting users, remote access services, and judicial systems. The tribunal network consists of firewalls, VLAN segmentation, Layer 2 switches, and internal connectivity between different court departments, including Civil, Small Claims, Penal, and Prosecution. Although VLAN segmentation provides some separation, the absence of internal firewall controls between court departments introduces a significant risk. If an attacker compromises a Clerk laptop or user account, they may be able to move laterally through the internal network and potentially access systems belonging to other court departments. This is particularly significant because Penal Courts and Prosecution handle more sensitive information requiring stronger confidentiality protections.

As shown in Table 2.1, laptops, specialised software, and network infrastructure have a direct relationship with both identified primary assets. Laptops and specialised software directly support the creation, processing, storage, and management of Court Case Information and the activities required to maintain Court Proceedings and Deadlines. Network Infrastructure has a direct relationship because it enables communication between users and systems containing primary assets, while weaknesses in network segmentation could directly affect confidentiality, integrity, and availability. In contrast, VPN and User Credentials have an indirect relationship because they do not store or process primary assets themselves but provide the authentication and connectivity mechanisms required to access systems containing these assets.

The identification of these supporting assets is particularly important because the proposed remote-working solution changes the security boundary of the tribunal environment. Remote endpoints, authentication mechanisms, remote access technologies, and shared network infrastructure increase the number of potential attack paths. A compromise of a supporting asset could allow attackers to access court information, modify judicial records, disrupt court operations, or affect legally required deadlines [2][4].

The analysis indicates that all five supporting assets are essential to maintaining the confidentiality, integrity, and availability of the Smart Court environment. Laptops and specialised software directly support judicial processing activities, while VPN, User Credentials, and Network Infrastructure provide the connectivity, authentication, and communication mechanisms required for secure operation. Consequently, compromise of any supporting asset could ultimately affect Court Case Information and Court Proceedings and Deadlines and will therefore be considered during the threat evaluation and risk assessment stages.

Table 1.2: Impact Assessment on Primary Assets

1.2 Impact Assessment on Primary Assets										
Primary Asset ID	Potential Compromise of C, I or A	Impact (see Table in Methodology)								Justification
		Personnel	Capacity	Performance	Economic	Branding	Regulatory	Environment	Overall impact	
Court Case Information	C	2	2	3	2	4	4	1	4	Disclosure of court records and legal documents could expose sensitive information, damage public confidence, and lead to breaches of privacy and data protection requirements.
	I	3	3	4	3	4	4	1	4	Unauthorised modification of case records or judicial decisions could affect the accuracy of proceedings, potentially resulting in appeals, retrials, and loss of trust in court outcomes.
	A	2	4	4	2	3	4	1	4	Unavailability of case information could delay hearings and court operations, reduce efficiency and potentially affect compliance with legal deadlines.
Court Proceedings and Deadlines	C	1	1	2	1	2	2	1	2	Court proceedings are generally intended to become public, making confidentiality less critical. However, premature disclosure may affect some ongoing cases.
	I	3	4	4	3	4	4	1	4	Incorrect procedural information or altered deadlines could disrupt proceedings, require corrective action, and negatively affect the fairness of legal processes.
	A	3	4	4	3	4	4	1	4	Delays or interruptions to court proceedings could affect case resolution times, create backlogs, and reduce confidence in the justice system.

Table 2.1: Linkage of Support Assets with Primary Assets

Supporting Assets		Linkage with Primary Assets		Justification
ID	Description	Court Case Information	Court Proceedings and Deadlines	
Laptops	Court-managed Windows 11 laptops issued to clerks for remote access to court	Direct	Direct	Clerks use laptops to access, view, update, and manage court case information and to carry out activities required for processing cases within required deadlines.

	systems and the completion of case-related administrative tasks.			
Specialised software	Judicial case management software used by clerks to manage case records, legal documents, workflow states, schedules, and judicial decisions.	Direct	Direct	The software directly stores, processes, and manages court case information while supporting the workflows and scheduling activities required to manage proceedings and deadlines.
VPN	The Virtual Private Network (Juniper Junos Pulse) used to provide secure remote connectivity between clerks' laptops and the court's internal systems.	Indirect	Indirect	The VPN enables secure access to systems containing the primary assets but does not itself store or process court case information or proceedings data.
User Credentials	Unique usernames and passwords assigned to each clerk for authentication and authorisation when accessing court systems remotely.	Indirect	Indirect	User credentials control access to supporting assets and systems that contain or manage the primary assets but do not directly store or process them.
Network Infrastructure	Tribunal network architecture consisting of firewalls, VLAN segmentation, Layer 2 switches, and internal connectivity between court systems, including Civil, Small Claims, Penal, and Prosecution departments.	Direct	Direct	The network infrastructure provides the communication path between users, remote access services, and judicial systems containing court case information. The absence of internal firewall separation between court departments creates a risk of lateral movement, where compromise of one court environment could expose sensitive information from other departments or disrupt judicial proceedings.

3.3 Threat Evaluation (Step 3)

At Step 3, threats affecting the supporting assets identified in Step 2 were evaluated (see Tables 3.1 and 3.2). Unlike a traditional court environment, the Smart Court solution introduces remote working capabilities, allowing judges and clerks to access court systems from outside the tribunal building through court-managed laptops and a VPN connection. This expands the security boundary of the tribunal and introduces additional attack paths involving endpoint devices, authentication mechanisms, remote access services, and shared network infrastructure. The identified threats are based on the specific characteristics of the Smart Court architecture and include malware infection, credential compromise, ransomware, unauthorised access, remote management software misuse, and network-based lateral movement.

For court-managed laptops, key threats include laptop theft or loss, malware infection, unauthorised access, and household access to devices. Because laptops are used outside the controlled environment of the tribunal, they may be exposed to physical theft, unauthorised viewing, or access by other individuals in the home environment. Furthermore, users may unintentionally download malicious software or open malicious attachments, allowing attackers to compromise the endpoint and potentially access court systems. These threats can affect the confidentiality, integrity, and availability of Court Case Information and may disrupt Court Proceedings and Deadlines by preventing access to required information or enabling unauthorised changes to case-related data.

For specialised judicial software, threats include ransomware attacks, data corruption, unauthorised modification of records, and misuse of remote management software. The judicial software is central to managing case folders, legal documents, workflow states, and procedural information. A successful attack against this supporting asset could modify case information, prevent access to records, or disrupt judicial processes. In particular, compromise of remote management software could provide attackers with additional access capabilities and increase the potential impact on court operations.

User credentials represent another important threat area because access to court systems depends on individual clerk accounts. The case study states that each clerk has a unique account and that sharing passwords can allow another person to impersonate the account owner. Therefore, password sharing, phishing, password reuse, and credential theft may allow attackers or unauthorised users to access court systems using legitimate credentials. Such compromise could result in unauthorised access to Court Case Information, modification of records, or disruption of court activities.

The VPN introduces additional risks because remote workers rely on it to connect from external locations to internal tribunal resources. A compromise of VPN credentials could allow an attacker to impersonate a legitimate clerk and access systems containing court information. Additionally, a VPN outage could prevent authorised users from completing required judicial tasks within legally defined deadlines. Although the VPN provides secure remote connectivity, its compromise directly affects the availability and confidentiality of remote court operations.

Network Infrastructure was identified as a critical supporting asset because it provides communication between court departments, remote access services, and internal judicial systems. The case study states that the tribunal network uses VLAN segmentation but also indicates that there is no internal firewall separating different court environments. This creates a significant risk of lateral movement between departments. For example, if a Civil Court clerk laptop or account is compromised, an attacker may be able to move through the internal network and attempt to access systems belonging to the Penal Court or Prosecution, where confidentiality requirements are significantly higher. Although VLAN separation provides some level of isolation, Layer 2 switching and the absence of internal firewall controls reduce the effectiveness of network segmentation.

The evaluated threats directly affect the primary assets: Court Case Information and Court Proceedings and Deadlines. Confidentiality risks mainly arise from credential compromise, laptop compromise, and network lateral movement, particularly because unauthorised access could expose sensitive case information or restricted judicial material. Integrity risks arise from threats such as unauthorised modification of records, compromised user accounts, and insufficient network separation, which could allow incorrect or malicious changes to case information. Availability risks are associated with ransomware, system outages, malware infection, and disruption of remote access services, which could delay court proceedings and prevent compliance with legally required deadlines.

The threat scenarios were derived from the specific characteristics of the Smart Court environment and evaluated using the SecRAM methodology, which links supporting assets, vulnerabilities, threats, impacts, and likelihood values to determine cybersecurity risks systematically [1]. The inherited impact values were based on the impact assessment established in Step 1, while reviewed impacts considered the mitigating circumstances available within the case study, such as restricted user privileges, account logging, existing VLAN separation, and court-managed devices. The analysis demonstrates how compromise of supporting assets can propagate to the primary assets and affect the confidentiality, integrity, and availability of court information and judicial operations.

Table 3.1: Threat Impact

Impact of Vulnerabilities & Threat Scenarios Evaluation																	
Supporting Asset	Threat	Vulnerability	Primary Assets												Inherited Impact	Reviewed Impact	Justification for attenuating circumstances
			Court Case Information			Court Proceedings and Deadlines			0			0					
			C	I	A	C	I	A	C	I	A	C	I	A			
Impact ⇨ (same as column "Overall impact" from Table 1.2)			4	4	4	2	4	4	0	0	0	0	0	0	MAX	<=	
Laptop	Theft of remote-working Clerk laptop	Device used outside secure court premises	Y		Y			Y							4	3	Devices are protected by Windows authentication, remote lock capabilities, and court IT support procedures.
	Malware infection	User downloads malicious software or opens malicious attachments	Y	Y	Y		Y	Y							4	3	Endpoint protection software and regular updates reduce the impact of malware incidents.
	Unauthorised access	Weak device security or unattended logged-in sessions	Y	Y			Y								4	4	Password policies and automatic screen locking reduce exposure.
	Household Access to Device	Devices used in shared home environments	Y	Y			Y								4	3	Windows authentication, screen-lock policies, and user accountability reduce the likelihood that household members can access or alter court information.
Specialised Software	Ransomware attack	Unpatched software vulnerabilities			Y		Y	Y							4	3	System backups exist, but disruption to court operations may still be significant until recovery is completed.
	Data corruption	Software defect or		Y	Y		Y	Y							4	3	Database backups and audit logs enable restoration of records.

		database error															
	Unauthorised modification of records	Excessive user privileges		Y			Y								4	3	Role-based access controls and audit trails help detect and reverse unauthorised changes.
	Remote Management Software Misuse	Remote administration tools exposed to compromise	Y	Y	Y		Y	Y							4	3	Administrative actions are logged and monitored, while access to remote management tools is restricted to authorised IT personnel, reducing the overall impact.
VPN	VPN credential theft	Weak passwords or credential reuse	Y												4	3	VPN monitoring and account lockout mechanisms reduce risk.
	VPN service outage	Technical failure or denial-of-service attack			Y			Y							4	3	Alternative access arrangements and IT recovery procedures minimise downtime.
User Credentials	Phishing attack	Lack of user awareness	Y	Y			Y								4	3	Security awareness training and email filtering reduce successful phishing attempts.
	Password Sharing	Clerks share passwords through external messaging applications	Y	Y			Y								4	3	Access logs, account monitoring, and supervisory reviews help detect suspicious account activity and limit the potential impact of credential misuse.
	Password reuse	Users reusing passwords across systems	Y	Y			Y								4	3	Password policies and periodic password changes reduce exploitation opportunities.

	Credential theft	Credentials exposed through malware or social engineering	Y	Y			Y								4	3	Monitoring, account lockouts, and incident response procedures limit damage.
Network Infrastructure	Lateral movement between court departments	No internal firewall separation between Civil, Small Claims, Penal, and Prosecution networks; VLAN segmentation relies partly on Layer 2 switching	Y	Y	Y	Y	Y	Y							4	3	The lack of internal firewall segmentation means that compromise of one court environment could allow attackers to move laterally into other departments. This may expose confidential case information, allow unauthorised changes to records, and disrupt judicial activities. Existing VLAN separation reduces likelihood, but it does not fully prevent movement between connected environments.
	Unauthorised access to tribunal systems through compromised network services	Over-reliance on global firewall protection without internal filtering between court environments	Y	Y	Y	Y	Y	Y							4	3	The global firewall protects the tribunal from external threats, but it does not provide separation between internal court departments. A compromised authorised device could therefore access systems belonging to other courts. Existing VLAN configuration and network monitoring reduce the potential impact.

Table 3.2: Threat likelihood.

Likelihood Assessment on Supporting Assets											
Supporting Asset (same as specified in step 2)	Threat	Vulnerability	Likelihood Areas (see Table in the Methodology)								
			Skills	Means	Opportunity	Profit	Attention	Impunity	Detection	Overall	Justification
Laptops	Laptop theft or loss	Device used outside secure court premises	1	2	4	2	3	2	4	3	Laptops are frequently used remotely, creating opportunities for theft or loss, although physical security controls reduce likelihood.
	Malware infection	User downloads malicious software or opens malicious attachments	2	3	5	3	3	3	3	4	Malware campaigns are common and require limited technical skill to execute.
	Unauthorised access	Weak device security or unattended sessions	2	2	4	2	2	2	3	3	Existing authentication controls and device management reduce the likelihood of successful exploitation.
	Household Access to Device	Devices used in shared home environments	1	1	4	2	2	4	3	3	Court-managed laptops are used in home environments where family members or visitors may gain physical access. The attack requires little technical skill and relies primarily on opportunity, making the likelihood moderate.
Specialised Software	Ransomware attack	Unpatched software vulnerabilities	4	4	3	5	4	3	2	4	Ransomware groups are highly capable and frequently target public-

											sector organisations for financial gain.
	Data corruption	Software defect or database error	2	2	2	1	1	5	3	2	Most corruption events are accidental rather than malicious and are mitigated through maintenance procedures.
	Unauthorised modification of records	Excessive user privileges	3	3	2	2	4	2	2	3	Requires authenticated access and sufficient permissions, reducing the opportunity for exploitation.
	Remote Management Software Misuse	Remote administration tools exposed to compromise	4	4	3	4	4	3	2	4	Compromising remote administration tools requires significant technical capability but could provide extensive access to court systems. The potential impact and value of such access make this an attractive target for attackers.
VPN	VPN credential compromise	Weak passwords or credential reuse	3	3	4	3	3	3	2	4	Credential attacks are common, particularly when users reuse passwords across systems.
	VPN service outage	Technical failure or denial-of-service attack	3	3	2	2	4	3	2	3	Service outages occur occasionally but are usually identified and resolved quickly.
User Credentials	Phishing attack	Lack of user awareness	2	2	5	3	2	4	3	4	Phishing is one of the most common attack methods and requires relatively little technical expertise.
	Password Sharing	Clerks share passwords through external	1	1	5	3	2	5	2	4	The case study assumes that some clerks share passwords through external messaging applications. This creates a direct opportunity for

		messaging applications									credential compromise and unauthorised access with minimal technical effort required by an attacker.
	Password reuse	Users reusing passwords across systems	2	2	3	3	2	4	2	3	Password reuse remains a common weakness despite organisational policies.
	Credential theft	Malware or social engineering	3	3	4	4	3	3	2	4	Stolen credentials are valuable and frequently used by attackers to gain unauthorised access.
Network Infrastructure	Lateral movement between court departments	No internal firewall separation between Civil, Small Claims, Penal, and Prosecution networks; VLAN segmentation relies partly on Layer 2 switching	4	4	3	5	4	3	2	4	Exploiting network segmentation weaknesses requires significant technical knowledge and access to an initial compromised device. However, the lack of internal firewall controls increases the opportunity for attackers to move between court environments. Access to sensitive judicial and prosecution information provides a high incentive, while network monitoring may detect unusual traffic patterns.
	Unauthorised access to tribunal systems through compromised network services	Over-reliance on global firewall protection without internal filtering between court environments	3	3	3	4	3	3	3	3	An attacker would need to compromise an authorised device or network service before exploiting internal connectivity. However, the absence of internal filtering increases the available attack surface. The potential access to court systems makes this attractive, while existing VLAN separation and monitoring reduce the

												likelihood of successful exploitation.
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3.4 Risk Evaluation (Step 4)

At Step 4, the impact and risk levels of the threats identified in Step 3 were evaluated (see Tables 4.1 and 4.2). The reviewed impact values from Table 3.1 were combined with the likelihood values from Table 3.2 using the SecRAM risk matrix shown in Table 4.1. This approach determines the overall risk level by considering both the potential consequences of a security incident and the likelihood that the threat scenario will occur [1].

The highest priority risks identified were Malware Infection affecting Laptops, Ransomware Attack affecting Specialised Software, Phishing Attack and Credential Theft affecting User Credentials, and Lateral Movement between Court Departments affecting Network Infrastructure. These threats were assessed as High risks because they combine significant reviewed impacts with likely or very likely occurrence levels. Successful exploitation of these threats could compromise the confidentiality, integrity, or availability of Court Case Information and Court Proceedings and Deadlines.

Malware infection and credential-related threats represent significant risks because remote-working clerks access court systems from outside the traditional security boundary of the tribunal. A compromised laptop or stolen credential could allow attackers to access judicial systems, modify case information, or disrupt court activities. The case study assumption that clerks may share passwords increases the likelihood of credential compromise because legitimate accounts could be misused to impersonate authorised users.

The ransomware risk affecting specialised judicial software was also assessed as High. Although recovery mechanisms such as backups may reduce the impact of permanent data loss, ransomware could still prevent clerks and judges from accessing case information and completing required judicial actions within legally mandated deadlines.

Network Infrastructure was identified as a critical risk area because of the absence of internal firewall separation between court departments. While VLAN segmentation provides some isolation, reliance on Layer 2 switching creates a potential pathway for lateral movement if an attacker compromises a device or account within one court environment. This is particularly significant because Civil and Small Claims Courts share infrastructure with more sensitive departments such as Penal Courts and Prosecution, where unauthorised disclosure could have severe consequences.

Although Laptop Theft or Loss was assessed as a Medium risk, it remains important because laptops are a primary access point to the Smart Court environment. A stolen or lost laptop could provide an attacker with access to court systems if additional security controls are bypassed. Other Medium-risk threats identified include Unauthorised Access, Unauthorised Modification of Records, Password Reuse, VPN Credential Compromise, VPN Service Outage, and Unauthorised Access through Compromised Network Services. These risks require mitigation because they could still affect court operations and statutory deadlines.

The evaluation demonstrates that threats affecting authentication mechanisms, endpoint devices, specialised judicial software, remote-access services, and network infrastructure can have cascading effects across the Smart Court environment. Successful exploitation may result in unauthorised disclosure of court information, modification of judicial records, disruption of court services, delays to proceedings, missed deadlines, and reduced public confidence in the Tribunal of Zaltbommel. Therefore, the High-risk threats and selected Medium-risk threats identified in Table 4.2 were prioritised for risk treatment in Step 5.

Table 4.1: Risk table.

Risk table with Reviewed Impact					
Likelihood	1. No impact, NA	2. Minor	3. Severe	4. Critical	5. Catastrophic
5. Certain	Low	High	High	High	High
4. Very likely	Low	Medium	High	High	High
3. Likely	Low	Low	Medium	High	High
2. Unlikely	Low	Low	Low	Medium	High
1. Very Unlikely	Low	Low	Low	Medium	Medium

Table 4.2: Risk evaluation

Risk evaluation					
	Threats (T 3.1)	Vulnerability (T3.1)	Reviewed Impact (T3.1)	Likelihood (T3.2)	Risk level (T4.1)
Laptop	Laptop Loss or Theft	Device used outside secure court premises	3	3	Medium
	Malware Infection	User downloads malicious software or opens malicious attachments	4	3	High
	Unauthorised Access	Weak device security or unattended sessions	4	2	Medium
Specialised Software	Ransomware Attack	Unpatched software vulnerabilities	3	4	High
	Data Corruption	Software defect or database error	3	2	Low

	Unauthorised Modification of Records	Excessive user privileges	3	3	Medium
User Credentials	Phishing Attack	Lack of user awareness	3	4	High
	Password Reuse	Users reusing passwords across systems	3	3	Medium
	Credential Theft	Malware or social engineering	4	4	High
VPN	VPN Credential Compromise	Weak passwords or credential reuse	3	3	Medium
	VPN Service Outage	Technical failure or denial-of-service attack	3	3	Medium
Network Infrastructure	Lateral movement between court departments	No internal firewall separation between Civil, Small Claims, Penal, and Prosecution networks; VLAN segmentation relies partly on Layer 2 switching	4	4	High
	Unauthorised access to tribunal systems through compromised network services	Over-reliance on global firewall protection without internal filtering between court environments	3	3	Medium

3.5. Risk Treatment (Step 5)

At Step 5, security controls were proposed to reduce the risks identified during the risk evaluation phase (see Table 5.1). The selected controls aim to reduce the likelihood of successful exploitation while maintaining the security requirements of the Smart Court environment. The proposed measures were based on recognised cybersecurity principles, including access control, vulnerability management, authentication protection, monitoring, backup and recovery, and network segmentation practices described in frameworks such as ISO/IEC 27001 and the NIST Cybersecurity Framework [2][5].

For laptop-related threats, controls were proposed to protect court-managed devices used outside the tribunal environment. Device encryption, remote wipe capability, and asset tracking reduce the consequences of laptop loss or theft by preventing unauthorised access to stored information. Endpoint Detection and Response (EDR), application control, and patch management were proposed to reduce the likelihood of malware infections by detecting malicious activity, restricting unauthorised software execution, and reducing exposure to known vulnerabilities. Multi-Factor Authentication (MFA), automatic lockout mechanisms, and enforced screen locking were proposed to mitigate unauthorised access to remote-working devices.

For specialised judicial software, centralised patch management, offline backups, and endpoint monitoring were proposed to reduce the risk of ransomware attacks. These controls improve the ability of the tribunal to recover from disruption while reducing the likelihood that vulnerabilities can be exploited. Database integrity controls and backup validation were proposed to reduce the impact of accidental corruption events, while least-privilege access controls and enhanced auditing were proposed to reduce the risk of unauthorised modification of court records.

User credential risks were addressed through phishing simulations, security awareness training, password managers, and Multi-Factor Authentication. These controls directly target High-risk threats identified during Step 4, particularly Phishing Attack and Credential Theft. Since the case study identifies password sharing between clerks as a possible risk, stronger authentication and monitoring controls are necessary to reduce the possibility of account impersonation and unauthorised activity.

For VPN-related threats, MFA and stronger password management controls were proposed to reduce the likelihood of VPN credential compromise. Redundant VPN infrastructure was proposed to improve availability and reduce disruption caused by VPN failures. These measures strengthen the remote-access capability required by clerks working outside the tribunal environment.

Network Infrastructure requires additional treatment because it represents a system-wide risk affecting multiple court departments. The case study states that VLAN segmentation exists but that there is no internal firewall separation between Civil, Small Claims, Penal, and Prosecution networks. To reduce the risk of lateral movement, internal firewall segmentation, stronger network access control rules, and monitoring of internal traffic between court environments were proposed. These controls reduce the likelihood that a compromised clerk device or account could move from one court environment into another. However, because the potential impact of compromise remains significant, particularly for sensitive Penal Court and Prosecution systems, the residual risk remains Medium rather than Low.

As shown in Table 5.1, the proposed controls mainly reduce the likelihood of successful attacks rather than eliminating their impact. Court Case Information and Court Proceedings and Deadlines remain critical assets, meaning that successful compromise would still have severe consequences. Most threats were reduced to Low residual risk through improved authentication, endpoint protection, monitoring, and recovery measures. However, Network Infrastructure risks remain at a Medium residual level because internal segmentation weaknesses require architectural improvements rather than only security monitoring controls.

The results demonstrate that the Smart Court remote-working solution can operate with an acceptable level of cybersecurity risk if the recommended controls are implemented and continuously maintained. However, ongoing monitoring, periodic review of access controls, and improvements to network segmentation remain essential to prevent compromise of sensitive judicial information and disruption of court proceedings.

Table 5.1: Risk Treatment and Calculation of Residual Risk for Supporting Assets

Risk Treatment and Calculation of Residual Risk for Supporting Assets										
Supporting Assets (T2.1)	Threats (T4.1)	Vulnerability (T4.1)	Pre-Controls	Post-Controls	Reviewed Impact (T4.1)	Likelihood (T4.2)	Residual Impact	Residual Likelihood	Residual Risk (T4.1)	Justification
Laptop	Laptop Theft or Loss	Device used outside secure court premises	Portable device used remotely	Device encryption, remote wipe, asset tracking	3	3	3	3	Low	Device encryption prevents unauthorised access to stored data, while remote wipe and asset tracking help protect and recover lost devices, reducing the risk of data exposure.
			Owner: IT Asset Management Team	Owner: IT Asset Management Team						
	Malware infection	Malicious downloads and attachments	Basic antivirus	EDR, application whitelisting, patch management	3	3	3	2	Low	EDR, application whitelisting, and regular patching reduce the chances of malicious software being executed and spreading across systems.
			Owner: IT Security Team	Owner: IT Security Team						
	Unauthorised Access	Weak device security	Password authentication	MFA, automatic lockout, screen-lock enforcement	4	2	4	1	Low	MFA and automatic lockout mechanisms make it significantly harder for attackers to gain access using stolen or guessed credentials.
			Owner: IT Security Team	Owner: IT Security Team						
Specialised Software	Ransomware Attack	Unpatched vulnerabilities	Periodic updates	Centralised patching, offline backups, EDR	3	4	3	2	Low	Centralised patching reduces exploitable vulnerabilities, while offline backups allow systems and data to be restored if an attack occurs.
			Owner: Application Support Team	Owner: Application Support Team						
	Data Corruption	Software defect or	Manual monitoring	Database integrity controls and backup validation	3	2	3	1	Low	Integrity controls and validated backups help identify corrupted data

		database error	Owner: Database Administrator (DBA)	Owner: Database Administrator (DBA)						quickly and ensure reliable recovery when needed.
	Unauthorised Modification	Excessive privileges	Existing permissions	Least-privilege model and enhanced auditing	3	3	3	2	Low	A least-privilege model limits unnecessary access rights, while auditing helps detect and investigate unauthorised changes.
			Owner: System Administrator	Owner: System Administrator						
User Credentials	Phishing Attack	Lack of Awareness	Basic awareness	Phishing simulations and security training	3	4	3	2	Low	Security training and phishing simulations improve user awareness, reducing the likelihood of employees falling for phishing attempts.
			Owner: Information Security Manager	Owner: Information Security Manager						
	Password Reuse	Reused passwords	Password policy	Password manager and MFA	3	3	3	2	Low	Password managers encourage unique passwords, and MFA provides additional protection if credentials are compromised.
			Owner: Information Security Team	Owner: Information Security Team						
	Credential Theft	Malware/social engineering	Reactive response	MFA, monitoring, anomaly detection	3	4	3	1	Low	MFA and monitoring tools reduce the likelihood that stolen credentials can be successfully used by attackers.
			Owner: Information Security Team	Owner: Information Security Team						
VPN	VPN Credential Compromise	Weak passwords	Password-only authentication	MFA and password manager enforcement	3	3	3	1	Low	MFA and stronger password practices make it much more difficult for attackers to gain VPN access using compromised credentials.
			Owner: Network Security Team	Owner: Network Security Team						
	VPN Service Outage	Technical failure	Single VPN gateway	Redundant VPN infrastructure	3	3	3	2	Low	Redundant VPN infrastructure provides

			Owner: Network Operations Team	Owner: Network Operations Team						backup connectivity, reducing service disruption and improving availability.
Network Infrastructure	Lateral movement between court departments	No internal firewall separation between Civil, Small Claims, Penal, and Prosecution networks; VLAN segmentation relies partly on Layer 2 switching	Global firewall and VLAN segmentation between external/public networks and tribunal systems	Implement internal firewall segmentation, network access control rules, stronger VLAN isolation, and east-west traffic monitoring between court departments	4	4	4	2	Medium	Existing VLAN separation provides some protection, but it does not fully prevent movement between court departments. Internal segmentation controls are required to reduce the possibility that a compromised Clerk laptop or account can reach sensitive Penal Court and Prosecution environments.
			Owner: Tribunal IT Management	Owner: Tribunal IT Management						
	Unauthorised access to tribunal systems through compromised network services	Over-reliance on global firewall protection without internal filtering between court environments	External firewall protecting the tribunal network boundary	Deploy internal filtering rules, intrusion detection, and continuous monitoring of internal network traffic	3	3	3	2	Medium	The global firewall reduces external attack exposure but does not prevent attacks originating from compromised authorised devices. Additional internal monitoring and filtering reduce the likelihood of unauthorised access to court systems.
			Owner: Tribunal IT Management	Owner: Tribunal IT Management						

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